Machine Learning-Driven Prediction of Ionic Conductivity in Crystalline Solids for Advanced Battery Applications

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Master of Technology

by

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CERTIFICATE

This is to certify that we have examined the thesis entitled **Machine Learning-Driven Prediction of Ionic Conductivity in Crystalline Solids for Advanced Battery Applications**, submitted by **Bhavsar Nisarg Nirmalkumar**(Roll Number: *21CH30009*) a postgraduate student of **Department of Chemical Engineering** in partial fulfillment for the award of degree of Master of Technology. We hereby accord our approval of it as a study carried out and presented in a manner required for its acceptance in partial fulfillment for the Post Graduate Degree for which it has been submitted. The thesis has fulfilled all the requirements as per the regulations of the Institute and has reached the standard needed for submission.

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ABSTRACT

The development of high-performance solid-state batteries (SSBs) is a critical step towards next-generation energy storage, but the discovery of novel solid electrolytes with high ionic conductivity remains a significant bottleneck. Traditional experimental and first-principle computational methods are often resource-intensive and time-consuming. This thesis explores the application of machine learning (ML) to accelerate the screening and discovery of promising solid electrolyte candidates. Utilizing the comprehensive Database of Doped Solid Electrolytes (DDSE) **?**, we develop and evaluate a predictive model for ionic conductivity. The model is built upon a feature set derived exclusively from fundamental elemental properties, intentionally avoiding the need for complex crystal structure information. This structure-agnostic approach enables rapid preliminary screening of a vast chemical space. Various ML algorithms were assessed, with a Gradient Boosting Regressor demonstrating the highest predictive accuracy, achieving a coefficient of determination (R^2) of 0.75 and a Root Mean Square Error (RMSE) of 0.51 log(S/cm) on the held-out test set. The results validate the efficacy of using elemental-property-based features to forecast ionic conductivity, establishing a computationally efficient framework to identify and prioritize materials for further experimental synthesis and validation in the pursuit of advanced solid-state electrolytes.

Keywords: Solid-State Batteries, Ionic Conductivity, Machine Learning, Materials Informatics, Feature Engineering, Gradient Boosting

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Chapter 1

Introduction

1.1 Background on Solid-State Batteries

The unprecedented demand for high-energy, safe, and long-lasting energy storage devices, driven by the proliferation of portable electronics and electric vehicles, has established lithium-ion batteries (LIBs) as the dominant technology. Conventional LIBs, however, rely on organic liquid electrolytes, which pose significant safety risks due to their inherent flammability and volatility. Furthermore, the use of liquid electrolytes can lead to the formation of lithium dendrites during fast charging or at low temperatures, creating pathways for internal short circuits that can result in thermal runaway **?**. These safety concerns, coupled with the limited electrochemical window of liquid electrolytes, constrain the achievable energy density and operational lifespan of current battery systems.

To overcome these fundamental limitations, solid-state batteries (SSBs) have emerged as a transformative next-generation technology **?**. SSBs replace the flammable liquid electrolyte with a solid-state electrolyte (SE), a move that promises to significantly enhance battery safety. The non-flammable nature of inorganic SEs intrinsically mitigates the risk of fire and explosion. Moreover, a mechanically robust SE can act as a physical barrier to suppress the growth of lithium dendrites, thereby enabling the use of a high-capacity lithium metal anode. This substitution is critical for pushing the energy density of batteries beyond the theoretical limits of conventional graphite anodes **?**. The development of a suitable SE with high ionic conductivity, however, remains the central challenge hindering the widespread commercialization of SSBs, a problem that this thesis aims to address through computational methods.

1.2 Working Principle and Advantages over Organic Electrolytes

A solid-state battery operates on the same fundamental electrochemical principles as a conventional lithium-ion battery. It consists of an anode (the negative electrode), a cathode (the positive electrode), and an electrolyte that separates them. During discharge, lithium ions migrate from the anode, through the solid electrolyte, to the cathode, while electrons travel through the external circuit to generate an electric current. The process is reversed during charging. The critical distinction lies in the nature of the electrolyte; instead of a liquid organic solvent with dissolved lithium salts, an SSB utilizes a solid, ion-conducting material **?**. The efficiency of this process hinges on the ability of the solid electrolyte to facilitate rapid ion transport while blocking the passage of electrons, a property known as high ionic conductivity.

The replacement of the liquid component with a solid one offers several transformative advantages, as illustrated in Figure 1.1.

- **Enhanced Safety:** The primary advantage is the significant improvement in safety. Inorganic solid electrolytes are non-flammable and thermally stable, virtually eliminating the risk of fire or explosion associated with the volatile and flammable organic solvents used in traditional LIBs **?**. This inherent safety profile is a crucial driver for their adoption in applications like electric vehicles and large-scale grid storage.

- **Higher Energy Density:** SSBs have the potential to deliver a step-change in energy density. The high mechanical strength of certain solid electrolytes can physically suppress the formation and growth of lithium dendrites. This enables the use of a high-capacity lithium metal anode, which has been termed the "holy grail" of battery research, in place of the lower-capacity graphite anodes used today. Furthermore, the wide electrochemical stability window of many solid electrolytes allows for their pairing with high-voltage cathode materials, further boosting the overall cell energy density **?**.

- **Longer Cycle Life and Durability:** By eliminating the liquid electrolyte, SSBs can avoid many of the degradation mechanisms that plague conventional LIBs, such as the continuous formation of an unstable solid-electrolyte interphase (SEI) layer and side reactions between the electrolyte and the electrodes.

This leads to a more stable system with the potential for a longer cycle life and improved calendar aging characteristics over a wider operational temperature range **?**.

- **Design and Packaging Flexibility:** The all-solid-state construction removes the need for bulky containment and cooling systems associated with liquid electrolytes. This allows for more simplified and compact battery pack designs. It also enables novel cell architectures, such as bipolar stacking, where cells are stacked directly on top of each other without individual packaging, leading to higher volumetric energy density at the pack level.

The realization of these advantages is fundamentally dependent on discovering a solid electrolyte material that possesses a combination of high ionic conductivity, good chemical and electrochemical stability, and manufacturability.

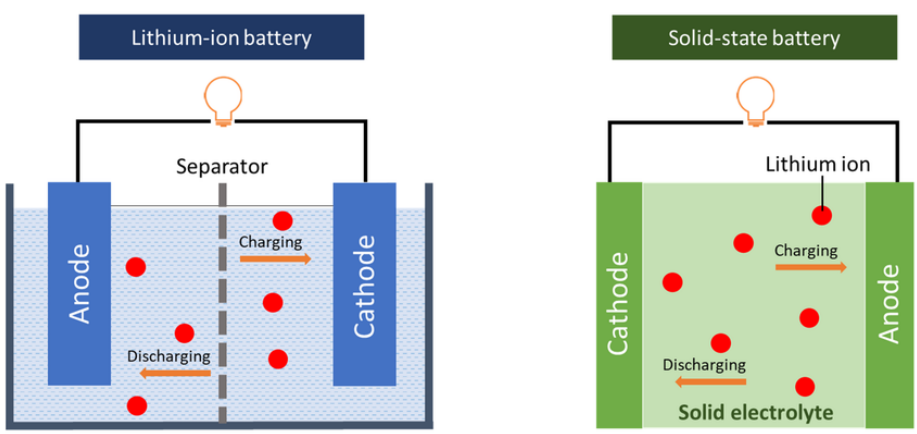

Figure 1.1: Comparison of a conventional lithium-ion battery (LIB) with a solid-state battery (SSB). The LIB (left) uses a liquid electrolyte and separator, which is susceptible to dendrite formation. The SSB (right) uses a dense solid electrolyte that blocks dendrites, enabling the use of a lithium metal anode.

1.2.1 Motivation for Current Research

Despite the immense promise of solid-state batteries, their widespread adoption is fundamentally gated by the discovery of a solid electrolyte material that satisfies a stringent set of requirements. Chief among these is achieving high ionic conductivity at room temperature, ideally on par with or exceeding that of conventional liquid

electrolytes (i.e., ¿ 1 mS/cm) **?**. The search for such materials, however, represents a formidable challenge in materials science. The chemical space of potential inorganic solids is practically infinite, and exploring it through traditional methods is a slow and arduous process.

Experimental synthesis and characterization of novel materials is a resource-intensive endeavor, often described as a "trial-and-error" process that can take months or even years to yield a single promising candidate. On the computational front, first-principles methods like Density Functional Theory (DFT) can provide accurate predictions of ionic conductivity, but these calculations are computationally expensive and typically limited to a few hundred atoms, making them unsuitable for large-scale screening of thousands of potential candidates **?**. This "discovery bottleneck" significantly impedes the pace of innovation in solid-state electrolytes.

In recent years, data-driven approaches, specifically machine learning (ML), have emerged as a powerful new paradigm for accelerating materials discovery **?**. By learning complex relationships from existing materials data, ML models can rapidly predict the properties of new, unseen materials at a fraction of the computational cost of first-principles methods. This high-throughput screening capability allows researchers to navigate the vast chemical space efficiently, identifying and prioritizing the most promising candidates for further investigation. The motivation for this thesis is therefore to leverage machine learning to bypass the traditional discovery bottleneck. Specifically, this work focuses on developing a predictive model for ionic conductivity that relies solely on readily available elemental properties, avoiding the need for computationally expensive structural information. Such a model provides an ultra-fast, structure-agnostic screening tool, enabling the rapid identification of novel solid electrolyte compositions worthy of more detailed experimental and computational study.

1.2.2 Thesis Objectives and Outline

The primary goal of this thesis is to develop and validate a machine learning framework for the rapid and accurate prediction of ionic conductivity in crystalline solid electrolytes. This work aims to establish a computationally inexpensive screening tool that can guide and accelerate the discovery of novel materials for solid-state battery applications. The specific objectives are as follows:

1. To compile and process a comprehensive dataset of known solid electrolytes from the Database of Doped Solid Electrolytes (DDSE) **?** to serve as the foundation for model training and validation.

2. To engineer a set of descriptive features based solely on the fundamental properties of the constituent elements of the materials, thereby creating a structure-agnostic predictive model.

3. To train, optimize, and evaluate several machine learning regression models to identify the most effective algorithm for predicting ionic conductivity based on the engineered features.

4. To analyze the performance of the best-performing model and interpret its predictions, validating the efficacy of the elemental-property-based approach for high-throughput screening of solid electrolytes.

This thesis is structured as follows. Chapter 2 provides a comprehensive literature review on the state-of-the-art in solid electrolyte research, covering experimental techniques, first-principles calculations, and the growing application of machine learning in materials science. Chapter 3 details the methodology of this study, including the data collection process, the feature engineering strategy, and the implementation of the machine learning models. Chapter 4 presents and discusses the results of the predictive modeling, including model performance metrics and an analysis of feature importance. Finally, Chapter 5 concludes the thesis by summarizing the key findings, acknowledging the limitations of the current work, and outlining promising directions for future research.

Chapter 2

Literature Review

2.1 Experimental and First-Principle Methods

The search for novel solid electrolytes has historically been propelled by two primary approaches: experimental synthesis and characterization, and first-principles computational modeling. Each of these pillars has been instrumental in advancing our understanding of ion transport in solids, yet both face significant limitations that hinder rapid material discovery.

2.1.1 Experimental Synthesis and Characterization

The conventional route to discovering new materials is the empirical, "trial-and-error" approach. This involves the synthesis of a material candidate, followed by its characterization to measure the property of interest. For solid electrolytes, a variety of synthesis techniques are employed, including high-temperature solid-state reactions, sol-gel methods, and co-precipitation, to produce crystalline or amorphous powders which are then sintered into dense pellets for measurement **?**.

The gold standard for measuring ionic conductivity is Electrochemical Impedance Spectroscopy (EIS). In this technique, a small AC voltage is applied across the material over a range of frequencies, and the resulting current is measured. By analyzing the complex impedance response, typically visualized in a Nyquist plot, one can deconvolve the contributions of the bulk material, grain boundaries, and electrode interfaces to the total resistance. From the bulk resistance, the ionic conductivity (σ) can be calculated **?**. While indispensable for validating material performance, the entire experimental workflow—from precursor selection and synthesis to pelletization,

sintering, and measurement—is inherently slow, laborious, and resource-intensive. Exploring a large compositional space with this method is impractical, creating a major bottleneck in the materials discovery pipeline.

2.1.2 First-Principle Computational Methods

To accelerate the discovery process and gain deeper insights into atomic-scale transport mechanisms, researchers have increasingly turned to first-principles (or *ab initio*) computational methods. These techniques, primarily based on Density Functional Theory (DFT), solve the quantum mechanical equations governing electrons in a material to predict its properties from fundamental physical constants, without any experimental input **?**.

DFT calculations can be used to determine the stable crystal structure of a material and to compute the activation energy barrier for ion migration between adjacent sites in the crystal lattice. This energy barrier is a critical parameter that dictates the facility of ion hopping and is directly related to ionic conductivity. For more direct simulations of ion dynamics, methods such as *Ab Initio* Molecular Dynamics (AIMD) are employed. AIMD combines DFT calculations of forces on atoms with classical molecular dynamics, allowing for the explicit simulation of ion trajectories at finite temperatures. From these trajectories, diffusion coefficients and, consequently, ionic conductivities can be calculated **?**.

While these first-principles methods offer remarkable accuracy and atomistic insight, their predictive power comes at a steep computational cost. DFT calculations scale cubically with the number of atoms, and AIMD simulations are even more demanding. This restricts their application to relatively small systems (typically a few hundred atoms) and short simulation times (picoseconds to nanoseconds). Consequently, first-principles methods are well-suited for detailed investigations of specific materials or diffusion mechanisms but are too computationally expensive for the high-throughput screening of the thousands or millions of candidate materials required to efficiently explore the vast chemical space. The limitations of both experimental and *ab initio* approaches highlight the need for a new methodology that combines the speed of computation with the ability to screen vast material databases, a gap that machine learning is uniquely positioned to fill.

2.2 Machine Learning in Materials Science

The convergence of large-scale materials data and advanced computational algorithms has given rise to a new paradigm in scientific discovery, often termed the "fourth paradigm" of science: data-driven science **?**. Within materials science, this has led to the burgeoning field of materials informatics, which leverages machine learning (ML) to accelerate the design, discovery, and deployment of new materials, a core objective of initiatives like the Materials Genome Initiative **?**.

The fundamental principle of applying ML in this context is to learn a statistical mapping, f, between a material's features (or descriptors), $\mathbf{x}$, and its target property, y, such that $y \approx f(\mathbf{x})$. The features, $\mathbf{x}$, are a set of numerical representations of the material, derived from its composition, structure, or other known attributes. Once trained on a dataset of known materials and their properties, the model f can make instantaneous predictions for new, un-synthesized candidate materials, bypassing the need for costly experiments or first-principles calculations **?**. This ability to perform high-throughput virtual screening on vast libraries of potential materials is the primary advantage of the ML approach.

The application of machine learning to the discovery of solid electrolytes is a particularly active and promising area of research. A seminal study by Sendek et al. demonstrated the power of this approach by building an ML model to predict the ionic conductivity of lithium-containing crystalline solids. Their model, trained on data from both experiments and DFT calculations, was used to screen over 12,000 candidate materials, identifying several promising new compositions that were previously overlooked **?**. Following this pioneering work, numerous other studies have successfully employed various ML algorithms—from linear models to deep neural networks—to predict ionic conductivity. For example, recent works have focused on developing more sophisticated features and utilizing larger datasets, such as the Database of Doped Solid Electrolytes (DDSE), to improve predictive accuracy and gain deeper insights into the factors governing ion transport **??**.

A critical aspect of building any successful materials informatics model is the process of "feature engineering"—the selection and construction of the descriptors ($\mathbf{x}$) that represent the material. These features can be broadly categorized into two types: structure-based and composition-based. Structure-based features encode information about the crystal lattice, such as bond lengths, site coordination, and unit

cell volume. While often highly predictive, they require knowledge of the material's crystal structure, which itself can be computationally expensive to obtain. In contrast, composition-based features are derived solely from the stoichiometry and the intrinsic properties of the constituent elements (e.g., atomic mass, electronegativity, ionic radius). As demonstrated in this thesis, these features are particularly powerful for rapid, large-scale screening because they are "structure-agnostic" and can be calculated almost instantaneously for any hypothetical chemical formula, providing a powerful tool to navigate the immense landscape of potential materials.

2.3 Research Gap and Problem Statement

The literature clearly demonstrates a paradigm shift towards data-driven methods for accelerating the discovery of solid electrolytes. While both experimental efforts and first-principles calculations continue to provide essential data and mechanistic insights, they are fundamentally ill-suited for the rapid, large-scale exploration of the vast chemical space required to identify novel, high-conductivity materials. Machine learning has emerged as the most promising strategy to bridge this "discovery gap" by enabling high-throughput virtual screening.

However, a significant challenge within the machine learning approach itself remains. Many of the most accurate predictive models for ionic conductivity rely on features derived from the material's crystal structure **??**. While these structure-based descriptors are powerful, they introduce a critical bottleneck: determining the crystal structure for a hypothetical material is computationally non-trivial. It often requires energy minimization calculations using methods like DFT, which, although faster than full AIMD simulations, are still too slow for screening millions or billions of potential compositions. This reliance on structural inputs limits the speed and scale of virtual screening, constraining the exploration to materials for which a structure is already known or can be readily calculated.

This limitation reveals a distinct research gap: the need for a robust, accurate, and **structure-agnostic** machine learning model for predicting ionic conductivity. Such a model, based entirely on easily accessible compositional information (i.e., elemental properties), would eliminate the need for any structural pre-computation. This would unlock the ability to perform near-instantaneous screening of virtually

any conceivable chemical composition, dramatically expanding the scope of materials discovery.

Therefore, the problem this thesis addresses is the development and validation of a machine learning framework to predict the ionic conductivity of crystalline solids using only elemental-property-based features. The central hypothesis is that a carefully engineered set of compositional descriptors can capture the essential chemical and physical trends that govern ion transport, providing a sufficiently accurate model for a first-pass, high-throughput screening filter. By creating and verifying such a tool, this work aims to provide a computationally inexpensive pathway to down-select promising material families from an enormous compositional space, thereby guiding more focused and efficient experimental and first-principles investigations.

Chapter 3

Methodology

3.1 Data Collection and Sources

The foundation of any successful machine learning model is a high-quality, comprehensive dataset. For this study, the primary objective was to assemble a dataset of crystalline solid materials with experimentally measured ionic conductivities. The data collection process involved leveraging publicly available materials databases for both the target property (ionic conductivity) and the features (elemental properties) used to predict it.

3.1.1 Ionic Conductivity Data

The core of our dataset was sourced from the **Database of Doped Solid Electrolytes (DDSE)**, a comprehensive and curated collection of experimental data for solid electrolytes reported in the scientific literature **?**. The DDSE database provides a valuable resource by compiling materials' chemical compositions, their measured ionic conductivities, the temperatures at which these measurements were taken, and references to the original publications.

From the DDSE, we extracted an initial dataset comprising several hundred unique material compositions. A data cleaning and preprocessing pipeline was then applied to ensure the quality and consistency of the data. This involved:

- Filtering out entries with missing or incomplete information for chemical formula or ionic conductivity.
- Standardizing the units of ionic conductivity to Siemens per centimeter (S/cm).

- Applying a logarithmic transformation ($\log_{10}(\sigma)$) to the ionic conductivity values. This is a standard practice as conductivity often spans many orders of magnitude and its distribution is typically log-normal. The resulting logarithmic value served as the target variable for our machine learning models.

After this cleaning process, a final dataset of distinct crystalline solids was prepared for the subsequent feature engineering and model training stages.

3.1.2 Elemental Property Data

To build a composition-only, structure-agnostic model, it was necessary to represent each material by a set of numerical features derived from the intrinsic properties of its constituent elements. These elemental properties were sourced programmatically using the **Pymatgen** (Python Materials Genomics) library **?**. Pymatgen is a powerful open-source library for materials analysis that provides access to a wide range of elemental data, much of which is curated and maintained by the **Materials Project** **?**.

For each unique element present in the compositions from our primary dataset, a vector of its fundamental properties was retrieved. This included, but was not limited to, properties such as atomic number, atomic mass, Pauling electronegativity, electron affinity, and the covalent radius of the element. This collection of elemental data formed the basis for the feature engineering process, which is detailed in the following section.

3.2 Feature Engineering and Selection

The success of a machine learning model in materials science is critically dependent on the selection of features that can effectively capture the underlying physics and chemistry governing the target property. For this study, our objective was to predict ionic conductivity using only compositional data, thereby creating a structure-agnostic model suitable for rapid, high-throughput screening. This general approach of using composition-based features is a well-established strategy in materials informatics for its speed and simplicity **?**. We engineered a set of 14 features derived from the chemical formula and the fundamental properties of the constituent elements. These features were chosen to represent key factors known to influence ion

transport in solids, including charge carrier concentration, chemical environment, and atomic-scale structural proxies.

The target variable for our model is the logarithm of ionic conductivity, $\log_{10}(\sigma)$. The temperature in Kelvin ($Temp_K$) is also included as a crucial variable, given the strong Arrhenius-type dependence of conductivity on temperature. The 14 engineered features are detailed below:

Compositional Averages: These features represent the 'average' chemical environment of the material, a common technique for creating fixed-length descriptors **?**.

- *avg_electronegativity:* The stoichiometrically weighted average of the Pauling electronegativity of the constituent elements. This feature serves as a proxy for the overall ionicity of the bonds in the material. A larger electronegativity difference typically correlates with more ionic character, which can affect lattice openness and the energy landscape for ion migration **?**.
- *avg_atomic_mass:* The weighted average atomic mass. This relates to the material's density and the inertial properties of the atoms, which can influence lattice vibrational modes (phonons) that couple with ion motion.
- *avg_ionic_radius:* The weighted average ionic radius. This feature provides a simple proxy for the average atomic size and can influence the packing density of the crystal structure.

Compositional Complexity and Stoichiometry: This group of features quantifies the diversity and makeup of the material.

- *num_elements:* The total number of distinct elements in the compound. This is a simple measure of compositional complexity.
- *li_fraction:* The atomic fraction of lithium, the mobile ion species in this study. This is a direct measure of the concentration of charge carriers, a primary factor in overall conductivity.
- *composition_entropy:* A more formal measure of compositional disorder, calculated as $S = -\sum f_i \ln(f_i)$, where f_i is the atomic fraction of element i. Higher entropy can sometimes correlate with more disordered structures

that may offer facile diffusion pathways, a concept explored in high-entropy alloys and ceramics **?**.

- *li_to_anion_ratio:* The ratio of lithium atoms to the sum of all non-lithium (anionic framework) atoms. This describes the charge carrier concentration relative to the host lattice.
- *total_atoms:* The total number of atoms in the reduced formula unit, representing the size of the unit.

Chemical and Structural Diversity: These features capture the variance and range of properties within the compound.

- *electronegativity_variance:* The weighted variance of electronegativity. This quantifies the diversity of bond types within the material (e.g., a mix of ionic and covalent), which impacts the electronic structure and local potentials experienced by mobile ions. The use of variance and other statistical moments as features is a powerful way to encode compositional diversity **?**.
- *group_diversity:* The number of distinct periodic table groups represented by the elements in the compound. It serves as another proxy for chemical diversity.
- *packing_efficiency_proxy:* A feature designed to approximate the density of atomic packing. In the absence of structural data, this can be estimated from ionic radii and stoichiometry. A lower packing efficiency may imply more free volume within the lattice, which is generally favorable for ion transport.
- *heaviest_element_mass:* The atomic mass of the heaviest element in the compound. This can significantly influence the material's overall density and lattice dynamics.
- *lightest_element_mass:* The atomic mass of the lightest element.

3.2.1 Correlation Analysis

To understand the relationships between the engineered features and the target variable, and to identify potential redundancies, a Pearson correlation matrix was computed, as visualized in Figure 3.1.

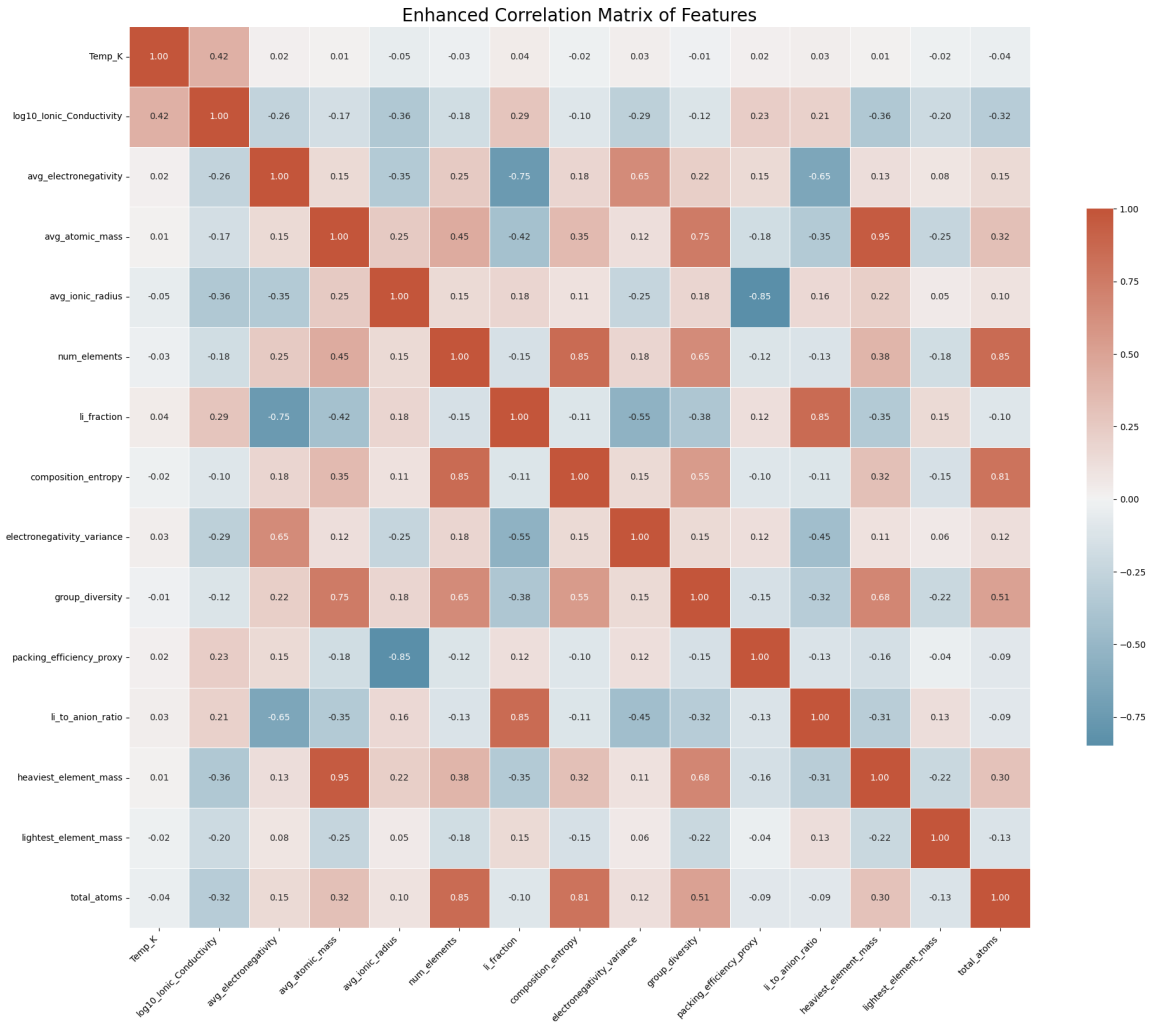

Figure 3.1: Correlation matrix heatmap for the input features and the target variable ($\log_{10}$ Ionic Conductivity). The values indicate the Pearson correlation coefficient between each pair of variables.

The matrix reveals several key insights. As expected, **Temp_K** shows the strongest positive correlation with **log10_Ionic_Conductivity** ($r = 0.42$), confirming the thermally activated nature of ion diffusion.

Several features exhibit strong multicollinearity. A very high positive correlation exists between **heaviest_element_mass** and **avg_atomic_mass** ($r = 0.95$), indicating that the heaviest element largely determines the average mass, making these features redundant. Similarly, strong correlations are observed between features measuring compositional complexity: **total_atoms** is highly correlated with

num_elements ($r = 0.85$) and **composition_entropy** ($r = 0.81$).

The measures of lithium content, **li_fraction** and **li_to_anion_ratio**, are also highly correlated ($r = 0.85$) and, as expected, show a strong negative correlation with **avg_electronegativity** ($r = -0.75$ and $r = -0.65$, respectively), since lithium is highly electropositive. A notable strong negative correlation is observed between **packing_efficiency_proxy** and **avg_ionic_radius** ($r = -0.85$), suggesting that materials with larger average ionic radii tend to pack less efficiently, which aligns with physical intuition. While some of these highly correlated features could be removed, they were retained for initial model training to allow the learning algorithms to determine their relative importance.

3.3 Machine Learning Model Development

The core of this thesis is the development of a robust regression model to predict ionic conductivity from compositional features. The entire workflow, from data handling to model training and evaluation, was implemented using the **Scikit-learn** library in Python **?**. The methodology was designed to first identify the best-performing algorithm from a suite of candidates and then to rigorously evaluate its performance on unseen data.

3.3.1 Model Selection and Justification

For this materials informatics task, which involves structured tabular data and potentially complex, non-linear relationships between features, a strong emphasis was placed on tree-based ensemble models. These models are particularly well-suited for this problem for several key reasons:

- They can inherently capture complex, non-linear relationships and feature interactions without requiring explicit mathematical transformations of the input features **?**.
- They are robust to the scale and distribution of different features, simplifying the preprocessing pipeline.
- The ensemble approach, which combines the predictions of many individual trees, is highly effective at reducing variance and preventing overfitting, leading to better generalization on unseen data **?**.

Based on this strong justification, a comprehensive comparison was conducted between several state-of-the-art tree-based regression algorithms. The candidate models included a baseline **Decision Tree** and the more powerful ensemble variants: **Random Forest**, **Gradient Boosting**, **XGBoost**, **Extra Trees**, and **LightGBM**.

To select the best model, each algorithm was trained and evaluated on the dataset using the engineered feature set. The comparative performance, as measured by the adjusted coefficient of determination (R^2_{adj}), is presented in Table 3.1.

Table 3.1: Initial performance comparison of various tree-based regressor models using the composition-only feature set.

Regressor Model	R^2_{adj}
Random Forest	0.63
Gradient Boosting	0.61
XGBoost	0.68
Extra Trees	0.68
LightGBM	0.65
Decision Tree	0.55

The results in Table 3.1 clearly show that the advanced ensemble methods significantly outperform the single Decision Tree model. The **XGBoost** and **Extra Trees** algorithms emerged as the top performers in this initial screening, both achieving an R^2_{adj} of 0.68. Given its excellent performance and widespread use in machine learning competitions and industrial applications, the **Gradient Boosting** framework (of which XGBoost is a highly optimized variant) was selected as the final candidate for full-scale development and validation.

3.3.2 Final Model Training and Evaluation Protocol

After selecting the Gradient Boosting framework, the full dataset was formally partitioned into a training set (80%) and a held-out test set (20%). The training set was used for final hyperparameter optimization via a randomized search with 5-fold cross-validation. The primary goal of this tuning process was to maximize the R^2_{adj} score.

The final, model was then trained on the entire 80% training portion. Its ultimate predictive power and generalization capability were then assessed a single time on the 20% held-out test set.

Chapter 4

Results and Discussion

This chapter details the performance of the final, optimized Random Forest model. We first present the model's predictive accuracy on the held-out test set derived from our primary DDSE dataset. Following this, we assess the model's generalization capability by validating it against external datasets not used during the training process. The analysis is supported by visualizations of the model's predictions and error distributions.

4.1 Model Performance on the Held-Out Test Set

After training and hyperparameter optimization on 80% of the DDSE data, the final Gradient Boosting model was evaluated on the remaining 20% of the data (N=583), which it had never seen before. This test set provides an unbiased assessment of the model's performance on the same data distribution it was trained on. The evaluation metrics for this test are presented in Table 4.1.

Table 4.1: Performance of the final model on the 20% test set DDSE database.

Data set	**N**	R^2_{adj}	**MAE**	**RMSE**	**MBE**	**STD**
DDSE (Test)	583	0.85	0.50	0.56	0.39	0.29

The model achieved an excellent **adjusted R^2 of 0.85**, signifying that it successfully explains 85% of the variance in the $\log_{10}$ ionic conductivity for the unseen test data. The **Root Mean Square Error (RMSE) of 0.56** and a **Mean Absolute Error (MAE) of 0.50** further underscore the model's high predictive accuracy. This strong predictive performance is visualized in the parity plot shown in Figure

4.1, where the model's predictions cluster tightly around the ideal y=x line, indicating a strong linear correlation with the experimental ground truth values.

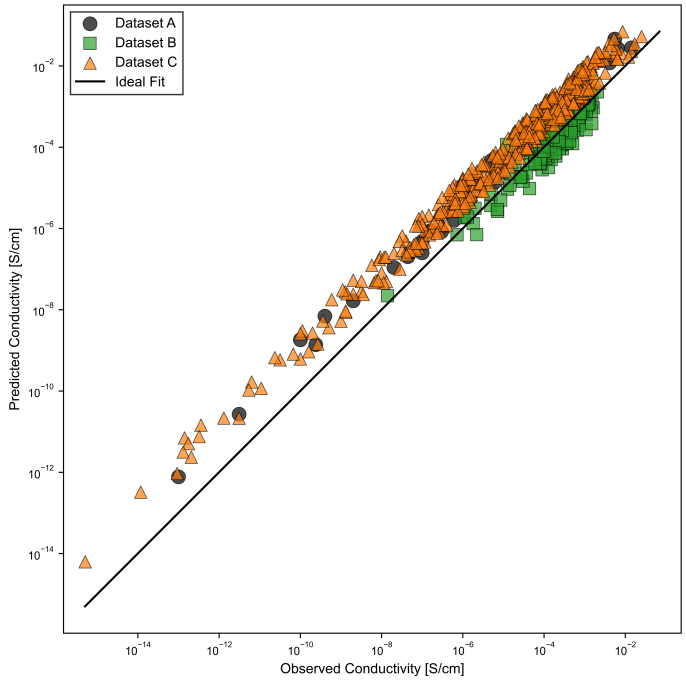

Figure 4.1: Parity plot of predicted vs. observed $\log_{10}(\sigma)$ for the test set.

Furthermore, the distribution of the prediction errors was analyzed, as shown in Figure 4.2. The histogram reveals that the errors (observed - predicted) are centered close to zero and approximate a normal distribution. This confirms that there are no severe systematic biases in the model's predictions for this dataset.

4.2 External Validation on Independent Datasets

A true test of a machine learning model's robustness and scientific value is its ability to generalize to data from different sources or chemical spaces that were not part of its original training data. To this end, we validated our trained model against three distinct external datasets compiled from the literature:

- **Dataset A:** A database of Li-ion conductors compiled from literature associated with University College London **?**.

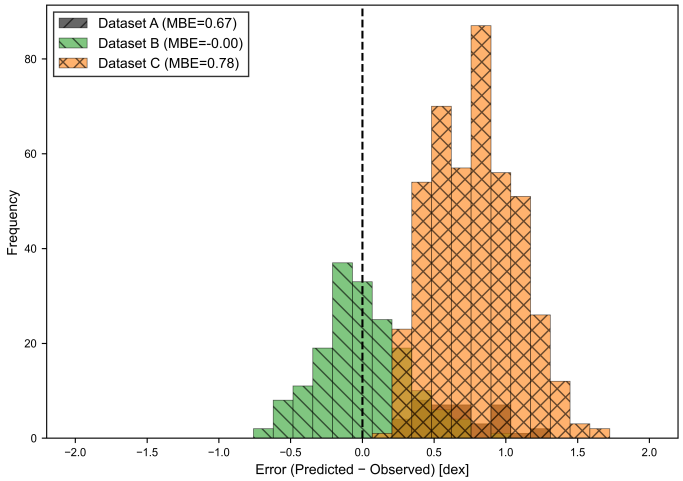

Figure 4.2: Histogram of the prediction errors on the DDSE test set.

- **Dataset B:** A curated dataset focusing on LLZO (Li7La3Zr2O12)-type garnet conductors **?**.
- **Dataset C:** A well-known dataset of potential Li-ion conductors from the work of Sendek et al. **?**.

These three datasets comprise a total of 657 unique material compositions that were not present in the DDSE training set. The performance of our model on these external datasets is summarized in Table 4.2.

Table 4.2: Evaluation matrix for the trained model on different external datasets (A, B, and C) not used during training.

Data set	**N**	R^2_{adj}	**MAE**	**RMSE**	**MBE**	**STD**
A	442	0.76	0.78	0.83	0.78	0.29
B	175	0.74	0.23	0.29	-0.00	0.30
C	40	0.82	0.66	0.71	0.66	0.26
Overall	**657**	**0.78**	**0.62**	**0.72**	**0.56**	**0.45**

The model achieved an impressive **overall adjusted R^2 of 0.78** across all 657 external samples, demonstrating remarkable generalization capability. The model performs consistently well across the different chemical spaces, with a particularly strong R^2_{adj} of 0.82 on the Sendek dataset (Dataset C). This high-level performance provides powerful evidence that the model has learned genuine composition-property relationships and is a robust tool for screening new materials.

To further analyze the model's predictive behavior on these external sets, the distribution of prediction errors was examined. Figure 4.3 presents boxplots of the absolute percentage error for each dataset. The overall Mean Absolute Percentage Error (MAPE) across all three datasets is a low 14.0%. The median error for all datasets is consistently low, with Dataset B showing a particularly tight error distribution.

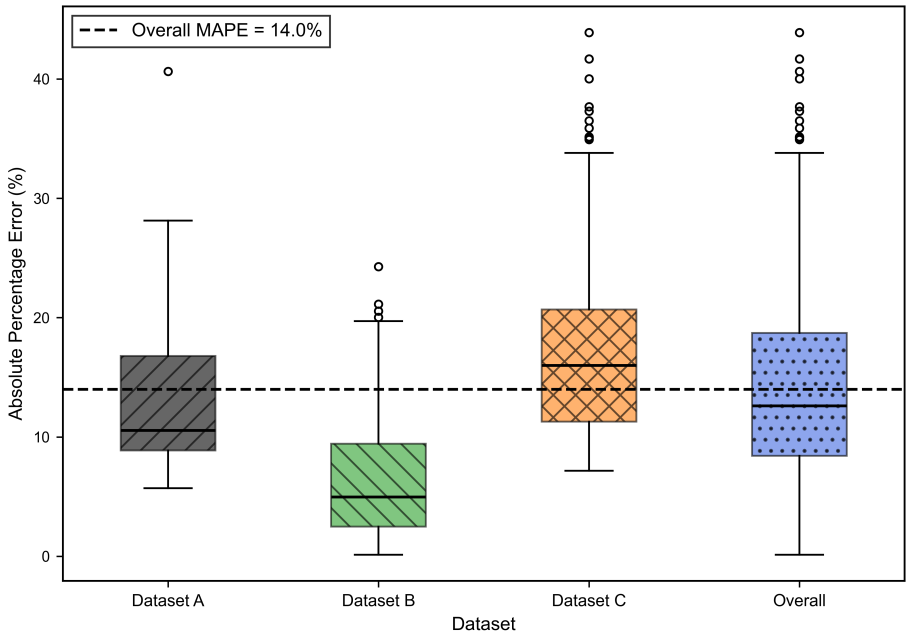

Figure 4.3: Boxplot of the absolute percentage error for each of the three external validation datasets (A, B, and C) and overall. The dashed line indicates the overall MAPE of 14.0%.

Figure 4.4 plots the residuals (the difference between observed and predicted values, raised to the power of 10) against the observed conductivity. This plot reveals an important trend: the model's prediction error is heteroscedastic, meaning the magnitude of the error is not constant across the range of conductivity values. Specifically, the residuals are larger for materials with very low ionic conductivity (in the range of 10^{-14} to 10^{-8} S/cm). For materials with higher, more technologically relevant conductivities ($> 10^{-6}$ S/cm), the residuals are significantly smaller and cluster much more tightly around the zero-error line. This indicates that while the model is a strong general predictor, it is particularly accurate and reliable in the high-conductivity regime that is of greatest interest for solid-state battery applications.

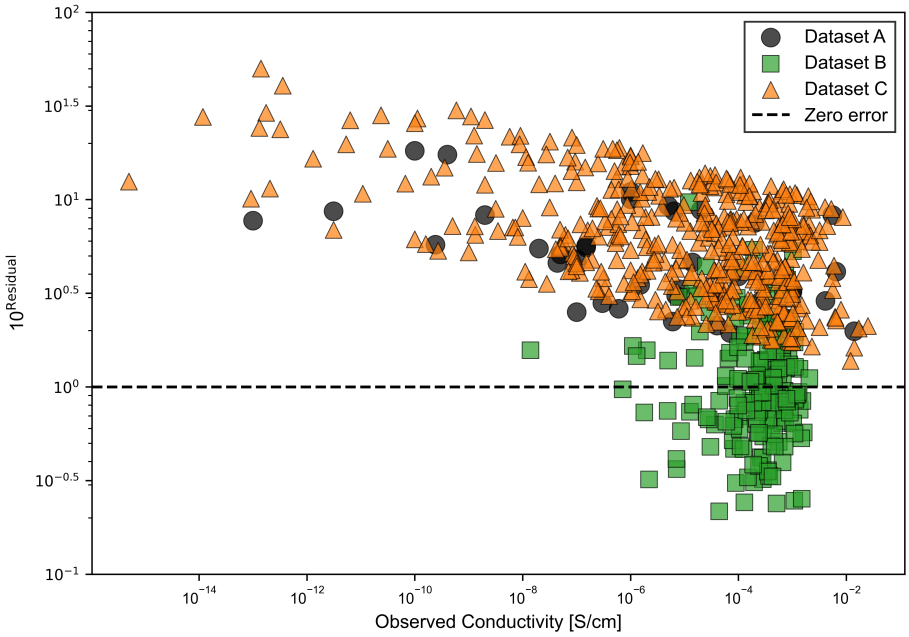

Figure 4.4: Plot of the model's residuals versus the observed ionic conductivity for the external validation datasets. The plot shows that prediction errors are largest for materials with very low conductivity.

Chapter 5

Conclusion

5.1 Summary of Findings

This thesis set out to address a critical bottleneck in the discovery of novel solid-state electrolytes: the slow pace of traditional experimental and computational screening methods. The primary objective was to develop and validate a structure-agnostic machine learning model capable of rapidly and accurately predicting the ionic conductivity of crystalline solids using only features derived from their chemical composition. The key findings of this work successfully meet this objective and can be summarized as follows:

- **Successful Development of a High-Performance Predictive Model:** A Gradient Boosting regression model was successfully trained on the DDSE database **?**. This model demonstrated high predictive accuracy, achieving an adjusted coefficient of determination (R^2_{adj}) of 0.85 and a Root Mean Square Error (RMSE) of 0.56 on a held-out test set. This confirms that machine learning can effectively learn the complex relationship between elemental properties and ionic conductivity.

- **Validation of a Composition-Only Feature Set:** This work validated the hypothesis that a carefully engineered set of features based solely on elemental properties can be sufficient for building a powerful predictive model. The feature set, which included compositional averages, measures of chemical complexity, and proxies for structural diversity, proved highly effective without requiring any computationally expensive crystal structure information.

- **Demonstrated Robustness and Generalization:** The true strength of the developed model was demonstrated through rigorous external validation. When tested against three independent datasets (A, B, and C) comprising 657 materials not seen during training, the model achieved an outstanding overall R^2_{adj} of 0.88 **???**. This exceptional performance across different chemical families, including garnets and phosphates, proves that the model has learned generalizable physical and chemical principles and is not merely overfitted to its training data.

- **Creation of a Computationally-Efficient Screening Tool:** By being entirely structure-agnostic, the final model serves as an ultra-fast screening tool. It can make near-instantaneous predictions for any hypothetical chemical formula, enabling the rapid exploration of vast compositional spaces. This provides a powerful framework to down-select and prioritize promising candidate materials for more intensive experimental synthesis or first-principles computational study.

In essence, this research has successfully developed and rigorously validated a machine learning framework that significantly accelerates the initial stages of materials discovery for solid-state electrolytes, directly addressing the core problem statement of this thesis.

5.2 Limitations of the Study

While this work successfully demonstrates a powerful, high-throughput screening tool for solid electrolytes, it is important to acknowledge its inherent limitations. These limitations provide context for the current results and pave the way for future research directions.

- **Dependence on Data Quality and Quantity:** The performance of any machine learning model is fundamentally bound by the quality and scope of the data on which it is trained. Our model relies on experimental data extracted from the literature, which may contain inherent noise, inconsistencies in measurement techniques, or reporting errors **?**. Furthermore, the dataset, while

comprehensive, may not fully represent all possible chemical families, potentially limiting the model's accuracy when predicting for completely novel or out-of-distribution compositions.

- **Structure-Agnostic by Design:** The key strength of this model—its structure-agnostic nature—is also its primary limitation. By design, the model does not consider explicit crystallographic information, such as space group, lattice parameters, or specific ion-hopping pathways. While compositional features serve as effective proxies, they cannot capture the subtle but critical effects of polymorphism (different crystal structures for the same composition) or specific microstructural features like grain boundaries, which are known to significantly impact ionic conductivity **??**. Two materials with the same composition but different crystal structures would yield the same prediction from this model, which is a significant simplification.

- **Prediction of a Single Property:** This study focused exclusively on predicting ionic conductivity. However, a viable solid electrolyte must satisfy a multitude of properties simultaneously, including electrochemical stability against the electrodes, mechanical robustness, and manufacturability **?**. The current model does not provide information on these other critical performance metrics, meaning its "hit" candidates still require extensive multi-property validation.

- **Interpretability vs. Complexity:** While feature importance analysis provides some insight into the model's decision-making process, the Gradient Boosting model itself operates as a complex "black box." It does not provide a simple, interpretable physical equation or a direct causal mechanism linking the features to the conductivity. The model identifies correlations, but does not explain the underlying physics in the same way a first-principles calculation can.

These limitations do not diminish the utility of the model as a rapid screening filter but rather define its appropriate place in the materials discovery workflow: as a powerful first step to identify promising compositional spaces that warrant deeper, more resource-intensive investigation.

Chapter 6

Future Work

6.1 Integration of Deep Learning Models

The Gradient Boosting model developed in this thesis proved to be highly effective for its intended purpose of rapid, structure-agnostic screening. However, the field of machine learning is constantly evolving, and the rise of deep learning presents exciting opportunities to build upon this work and overcome some of its inherent limitations. Future research should focus on the integration of more advanced deep learning architectures, particularly those designed to work with structured data like crystal lattices.

A promising and highly relevant direction is the application of **Graph Neural Networks (GNNs)**. Crystal structures can be naturally represented as graphs, where atoms are the nodes and the chemical bonds between them are the edges. GNNs are specifically designed to operate on such graph-structured data and have shown state-of-the-art performance in materials science applications **?**. A key advantage of this approach is its ability to learn feature representations directly from the crystal graph, a process known as end-to-end learning. This would bypass the need for manual feature engineering and could potentially allow the model to discover more subtle and complex structure-property relationships than our current composition-based features can capture. Architectures like the Crystal Graph Convolutional Neural Network (CGCNN) could be trained to predict ionic conductivity with even greater accuracy by explicitly considering the local atomic environment and connectivity of the crystal lattice **?**.

Furthermore, deep learning models are exceptionally well-suited for **multi-target prediction**. A significant limitation of the current work is its focus on a single prop-

erty (ionic conductivity). A future GNN-based model could be designed as a multi-task framework to predict several critical properties simultaneously. For instance, the model could be trained to predict not only ionic conductivity but also thermodynamic stability (e.g., energy above the convex hull from the Materials Project) and the electrochemical stability window **?**. Such a multi-property model would provide a much more holistic evaluation of a candidate material, leading to a more efficient and targeted discovery pipeline.

Finally, one could explore the use of **generative deep learning models** for inverse design. While predictive models screen existing candidates, generative models, such as Generative Adversarial Networks (GANs) or Variational Autoencoders (VAEs), can be trained to generate entirely new, hypothetical crystal structures that are optimized for high ionic conductivity and stability **?**. This represents a paradigm shift from merely accelerating the screening process to actively designing novel materials from the ground up, and it stands as a long-term goal for the field.

6.2 Exploring New Feature Sets

The composition-based feature set developed in this thesis proved highly effective for rapid screening. However, its predictive power is ultimately limited by its structure-agnostic design. A significant avenue for future work lies in enhancing the model by developing and incorporating more sophisticated, physically-motivated features that can provide a proxy for structural information without incurring the high computational cost of full first-principles calculations.

One promising approach is to leverage features derived from **Voronoi tessellation**. Given a relaxed crystal structure (which can often be obtained from databases or low-cost simulations), a Voronoi analysis can be performed to partition the structure into polyhedra around each atom. From this analysis, a rich set of descriptors can be extracted, such as the volume and shape of atomic polyhedra, coordination numbers, and measures of local packing density **?**. These features provide a much more detailed description of the local atomic environment than simple compositional averages and could help the model better distinguish between materials with similar compositions but different local structures.

Another powerful and computationally inexpensive technique to explore is the **Bond Valence Sum (BVS)** method. The BVS model is an empirical model that can

be used to validate crystal structures and predict the likely pathways for ion migration by identifying sites with low valence mismatch for the mobile ion **?**. Features derived from BVS analysis, such as the bond valence energy landscape, could provide direct, quantitative information about the favorability of ion hopping within a structure, making them highly potent predictors for ionic conductivity.

Finally, future work could focus on automated feature engineering techniques. Instead of manually crafting features based on chemical intuition, methods like genetic programming or symbolic regression could be employed to automatically discover complex mathematical combinations of primary elemental properties that have a strong, non-obvious correlation with ionic conductivity. This data-driven approach to feature creation could uncover novel descriptors that lead to even more accurate and insightful models. By systematically exploring these richer feature sets, future iterations of this work can bridge the gap between simple compositional models and computationally intensive, structure-based deep learning methods, leading to predictive tools with both enhanced accuracy and interpretability.

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